



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

<Name>

<Date>



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodology

- Data collected via SpaceX REST API and Wikipedia web scraping, then wrangled and labeled for binary classification (landed vs. did not land).
- EDA performed with SQL queries, scatter/bar/line charts, and interactive Folium maps and Plotly Dash dashboards.
- Four classification models trained with GridSearchCV (10-fold CV): Logistic Regression, SVM, Decision Tree, and KNN.

Key Results

- Landing success rose to 90% (2019), driven by booster evolution and operational experience.
- KSC LC-39A led all sites with a 76.9% success rate; orbit type strongly influenced outcomes (GTO ~50%, ES-L1/GEO/HEO/SSO at 100%).
- Decision Tree achieved 94.44% accuracy (best model), with zero missed landings and only one false positive.

Introduction

Background

- SpaceX advertises Falcon 9 launches at \$62M vs. \$165M+ for competitors, largely because they recover and reuse the first stage booster. Predicting whether a launch will successfully land its booster is key to estimating launch cost and competitiveness.

Research Questions

- Which factors (site, orbit, payload, booster version) most influence landing success?
- Has landing success improved over time, and why?
- Can we build a classification model to predict first-stage landing outcomes?

Section 1

Methodology

Methodology

Executive Summary

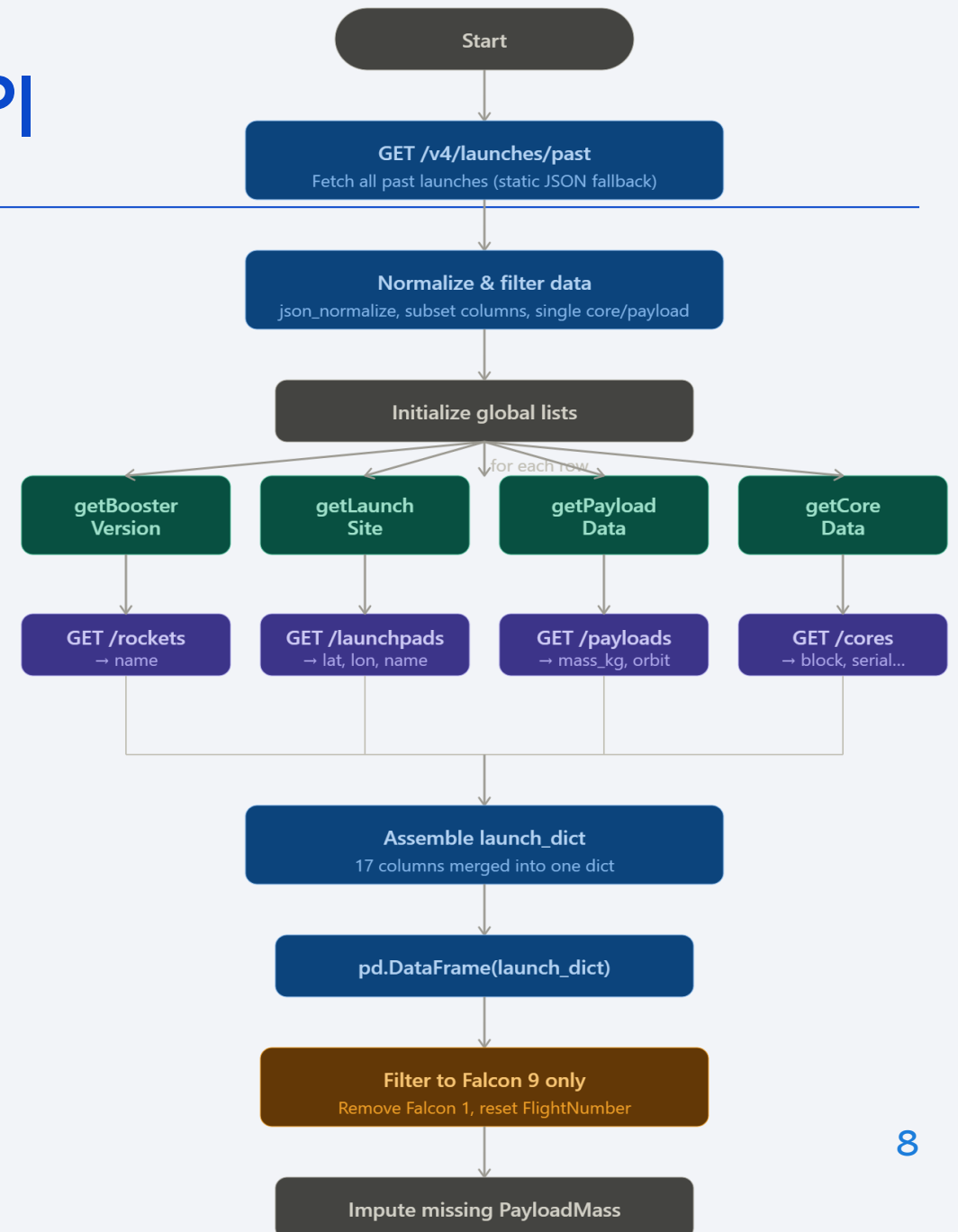
- Data collection methodology:
 - Data collected via SpaceX REST API and Wikipedia web scraping, then wrangled and labeled for binary classification (landed vs. did not land).
- Perform data wrangling
 - Nulls imputed, outcomes encoded into a binary label (1 = landed, 0 = failed), and features one-hot encoded for ML input
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Trained LR, SVM, Decision Tree, and KNN with GridSearchCV (10-fold CV); Decision Tree scored highest at 94.44%

Data Collection

- **API:** Queried the official SpaceX REST API across five endpoints — pulling structured JSON data for launches, rockets, launchpads, payloads, and cores, then merged into a single DataFrame.
- **Web Scraping:** Scraped Wikipedia's Falcon 9 launch history using BeautifulSoup, parsing raw HTML tables row by row and cleaning messy formatting before exporting to CSV.
- **Bottom line:** API = clean and direct. Scraping = flexible but messier.
- You need to present your data collection process use key phrases and flowcharts

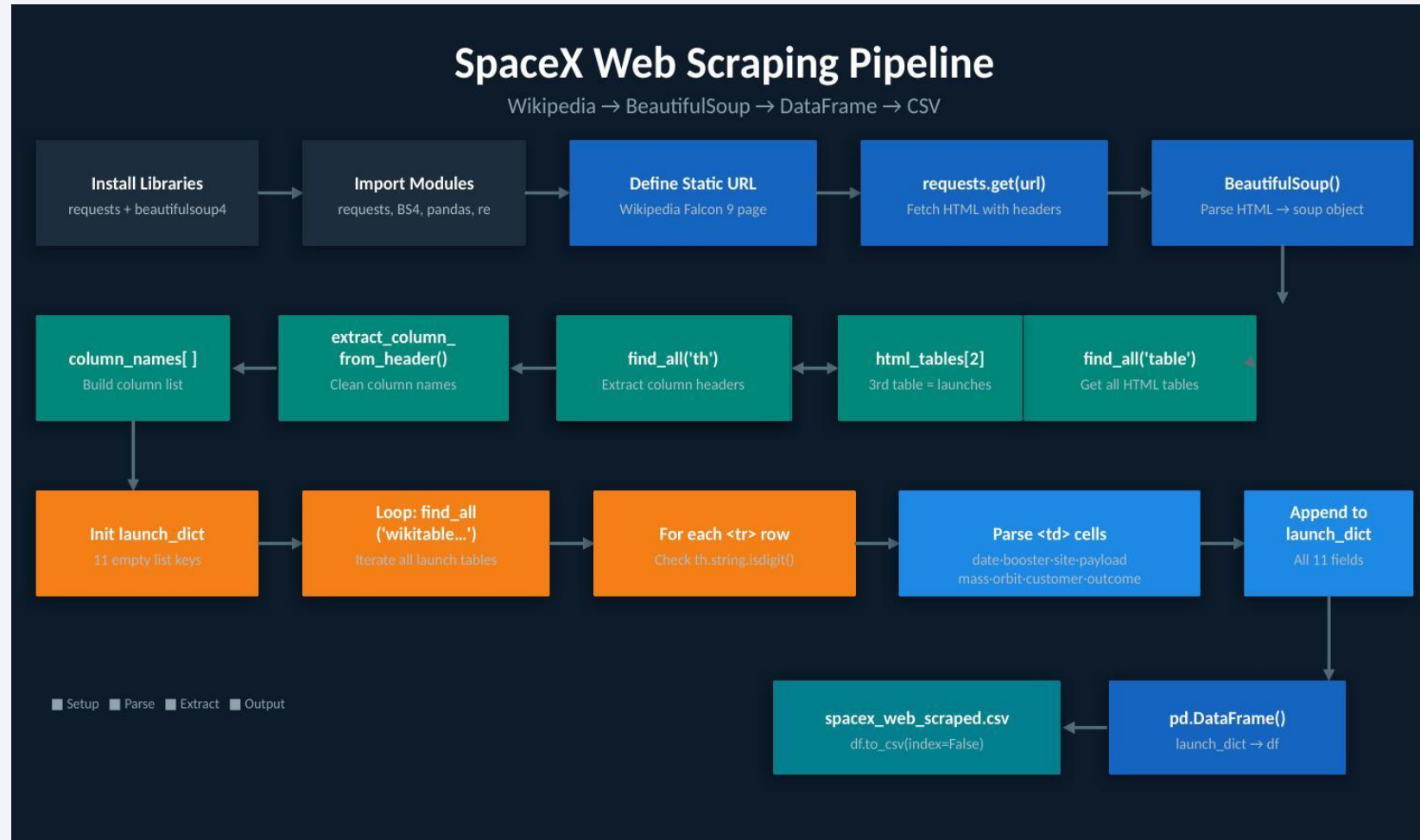
Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- [DataAnalytics-Science-Projects/Data Science Capstone project content/jupyter-labs-spacex-data-collection-api \(1\).ipynb](#) at main · [Cuboon/DataAnalytics-Science-Projects](#)



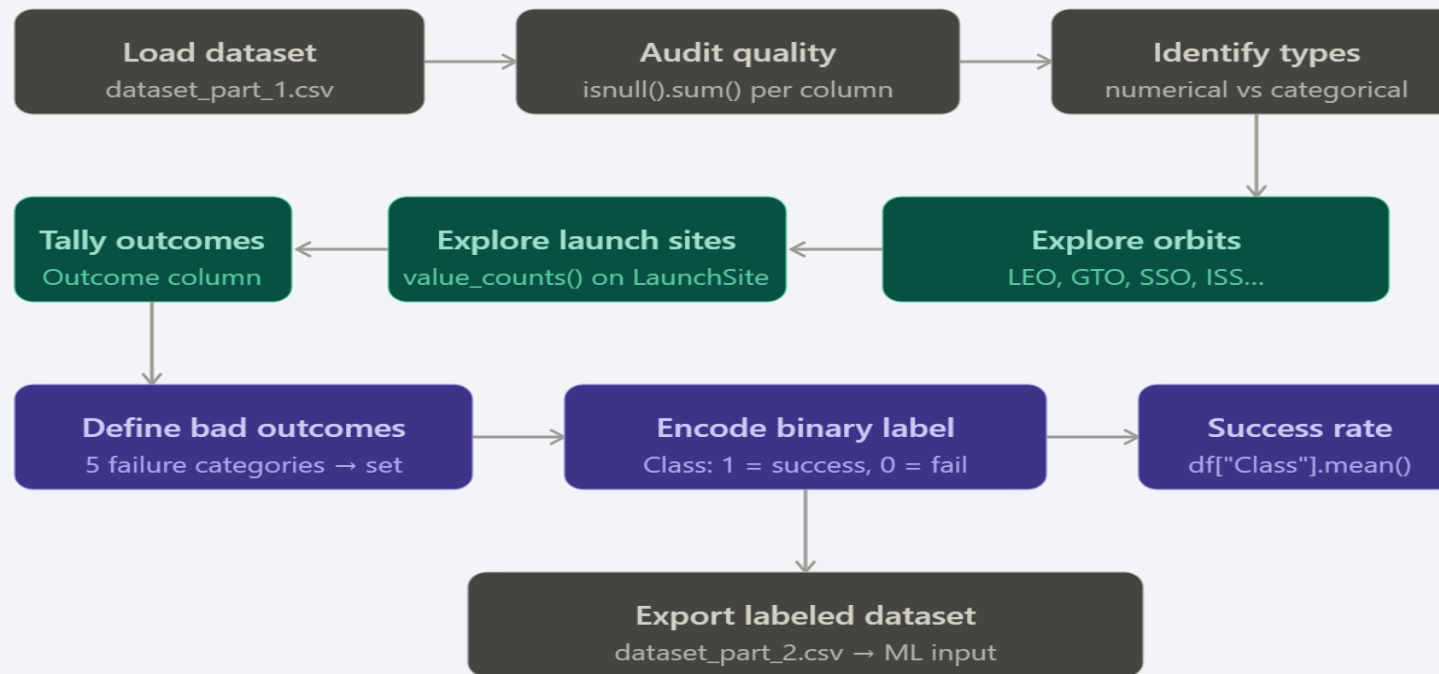
Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- [DataAnalytics-Science-Projects/Data Science Capstone project content/jupyter-labs-webscraping \(1\).ipynb at main · Cuboon/DataAnalytics-Science-Projects](#)



Data Wrangling

- Load dataset from IBM Cloud → inspect nulls → identify column types → explore launch sites → explore orbit types → analyze landing outcomes → encode binary label → export CSV
- You need to present your data wrangling process using key phrases and flowcharts



EDA with Data Visualization

- [DataAnalytics-Science-Projects/Data Science Capstone project content/edadataviz \(1\).ipynb at main · Cuboon/DataAnalytics-Science-Projects](#)

Chart	Variables	Why
Scatter	Flight No. vs Payload Mass	More flights = higher success; payload weight is a factor
Scatter	Flight No. vs Launch Site	Tracks site usage & success over time
Scatter	Payload Mass vs Launch Site	VAFB-SLC: no launches above 10,000 kg
Bar	Success Rate by Orbit	Compares landing outcomes across orbit types
Scatter	Flight No. vs Orbit Type	LEO improves with experience; GTO shows no trend
Scatter	Payload Mass vs Orbit Type	Polar/LEO/ISS handle heavy loads better than GTO
Line	Yearly Success Rate	Upward trend 2013–2020 confirms learning over time

EDA with SQL

- Unique launch sites — retrieved all distinct launch site names using DISTINCT
- Site filter — returned 5 records where launch site starts with 'CCA' using LIKE
- Total payload (NASA CRS) — summed all payload mass for NASA CRS booster launches using SUM
- Avg payload (F9 v1.1) — calculated average payload mass for booster version F9 v1.1 using AVG
- First successful ground landing — found the earliest success date on a ground pad using MIN
- Drone ship successes by payload — listed boosters with drone ship success and payload between 4,000–6,000 kg using BETWEEN
- Mission outcome counts — tallied total successful vs failed missions using GROUP BY + COUNT
- Max payload boosters — identified booster versions carrying the heaviest payload using a subquery with MAX
- 2015 drone ship failures — extracted month, booster, and site for all drone ship failures in 2015 using substr() for date parsing
- Ranked landing outcomes — ranked all landing outcome types between 2010–2017 by frequency using GROUP BY + ORDER BY DESC
- [DataAnalytics-Science-Projects/Data Science Capstone project content/jupyter-labs-eda-sql-coursera_sqlite \(1\).ipynb at main · Cuboon/DataAnalytics-Science-Projects](#)

Build an Interactive Map with Folium

- `folium.Circle` — placed a labeled circle at each of the 4 launch sites (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E) to visually pinpoint their geographic coordinates on the map.
 - `Marker with DivIcon` — added a text label next to each circle showing the site name, making the map readable without needing to click
 - `MarkerCluster` — grouped all individual launch records into clusters to avoid clutter, since many launches share the exact same coordinates; clusters expand on zoom
 - `Color-coded folium` — marked every launch outcome per site: green = success (Class 1), red = failure (Class 0); allows instant visual identification of high vs low success rate sites
 - `MousePosition plugin` — added a live coordinate display on hover so exact lat/lon of nearby features (coastline, highway, railway) could be read off the map and used in distance calculations
 - `PolyLine` — drew a straight line from the launch site to its closest coastline point to visually represent and measure proximity; the same approach was applied to nearest city, highway, and railway
 - `Icon Marker` — placed a text label at the coastline endpoint showing the calculated distance in km, computed using the Haversine formula
- Why geographic proximity matters: Launch sites are consistently near coastlines (for safety — debris falls into the ocean), near the equator (to benefit from Earth's rotation), but kept away from cities to minimize risk to populated areas.
- [DataAnalytics-Science-Projects/Data Science Capstone project content/lab jupyter launch site location.ipynb at main · Cuboon/DataAnalytics-Science-Projects](#)

Build a Dashboard with Plotly Dash

Plots/Graphs

- **Pie chart** (success-pie-chart) — shows total successful launches per site when "All Sites" is selected; switches to success vs failure ratio for a single site when one is chosen — lets you instantly compare which sites perform best and diagnose weak sites
- **Scatter chart** (success-payload-scatter-chart) — plots payload mass (kg) on the x-axis against launch outcome (0/1) on the y-axis, color-coded by booster version — reveals whether heavier payloads reduce success rates and which booster versions handle them best

Interactions

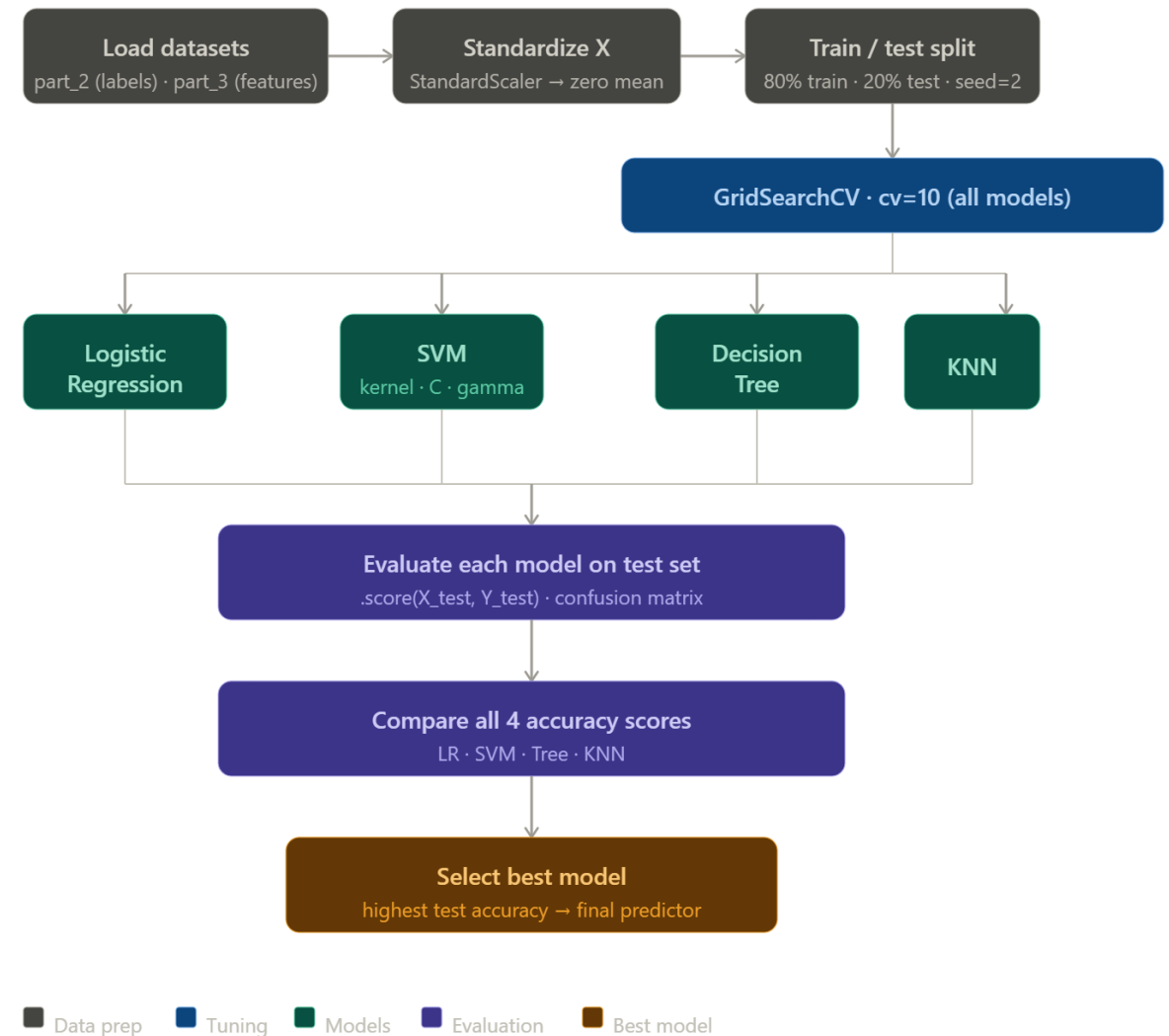
- **Dropdown** (site-dropdown) — filters both charts by launch site or shows all sites combined — allows focused analysis on a single site without rebuilding the chart manually
- **Range slider** (payload-slider) — sets a min/max payload window that filters the scatter chart in real time — lets you isolate specific payload ranges to spot patterns (e.g. do mid-weight payloads land more reliably than heavy ones?)

How they work together

- Dropdown + pie chart → identify best-performing sites
- Dropdown + slider + scatter → investigate *why* a site succeeds or fails by correlating payload weight and booster version with outcomes
- [DataAnalytics-Science-Projects/Data Science Capstone project content/spacex-dash-app \(1\).py at main · Cuboon/DataAnalytics-Science-Projects](#)

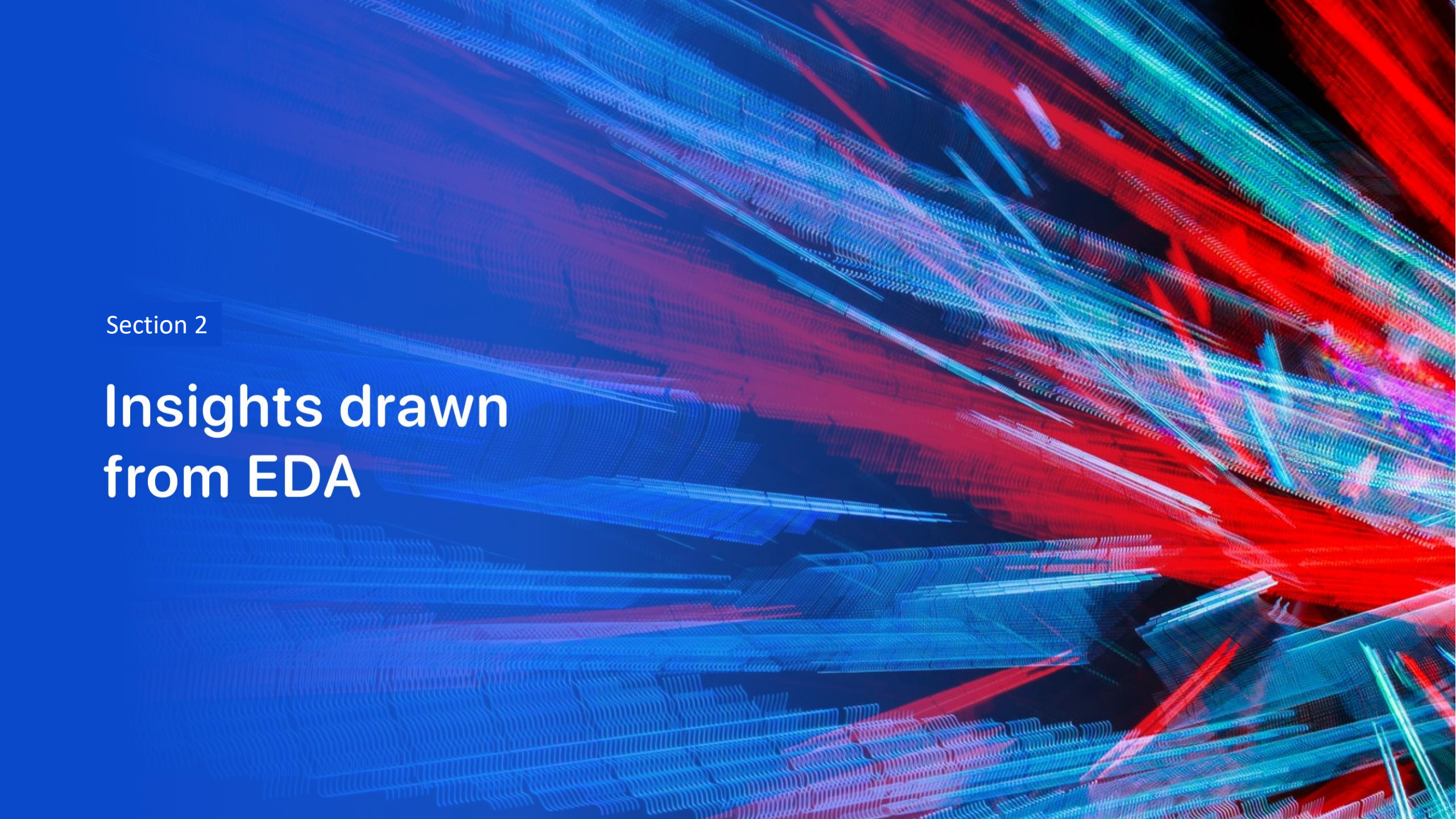
Predictive Analysis (Classification)

- Load data — dataset_part_2.csv (labels) + dataset_part_3.csv (one-hot encoded features)
- Extract target — Class column → NumPy array Y
- Standardize — StandardScaler applied to X (zero mean, unit variance)
- Split — 80% train / 20% test · random_state=2 · 18 test samples
- Tune & train x4 — GridSearchCV with cv=10 on all 4 models
- Evaluate — test accuracy + confusion matrix per model
Select best — compare all 4 scores → pick highest
- [DataAnalytics-Science-Projects/Data Science Capstone project content/SpaceX Machine Learning Prediction Part 5.ipynb at main · Cuboon/DataAnalytics-Science-Projects](#)



Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

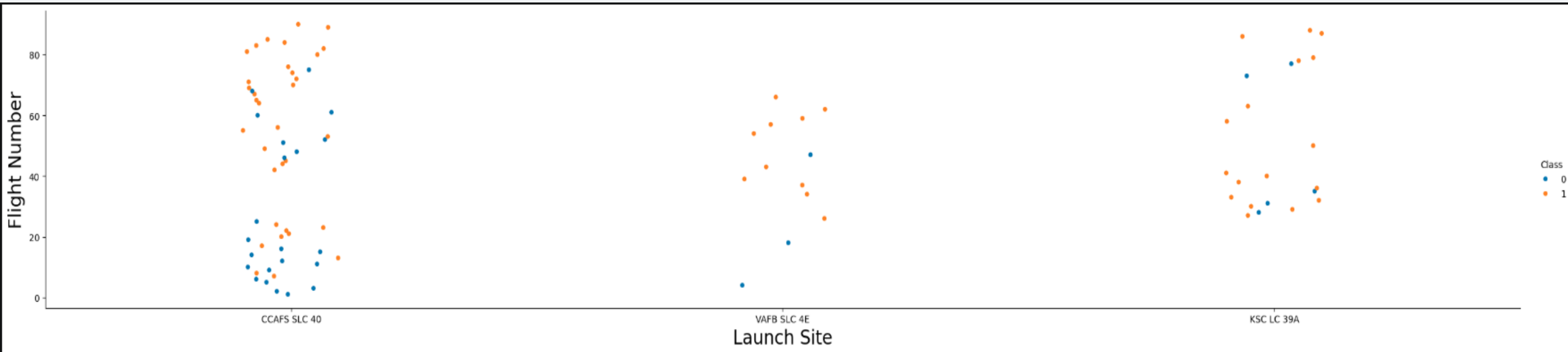


Section 2

Insights drawn from EDA

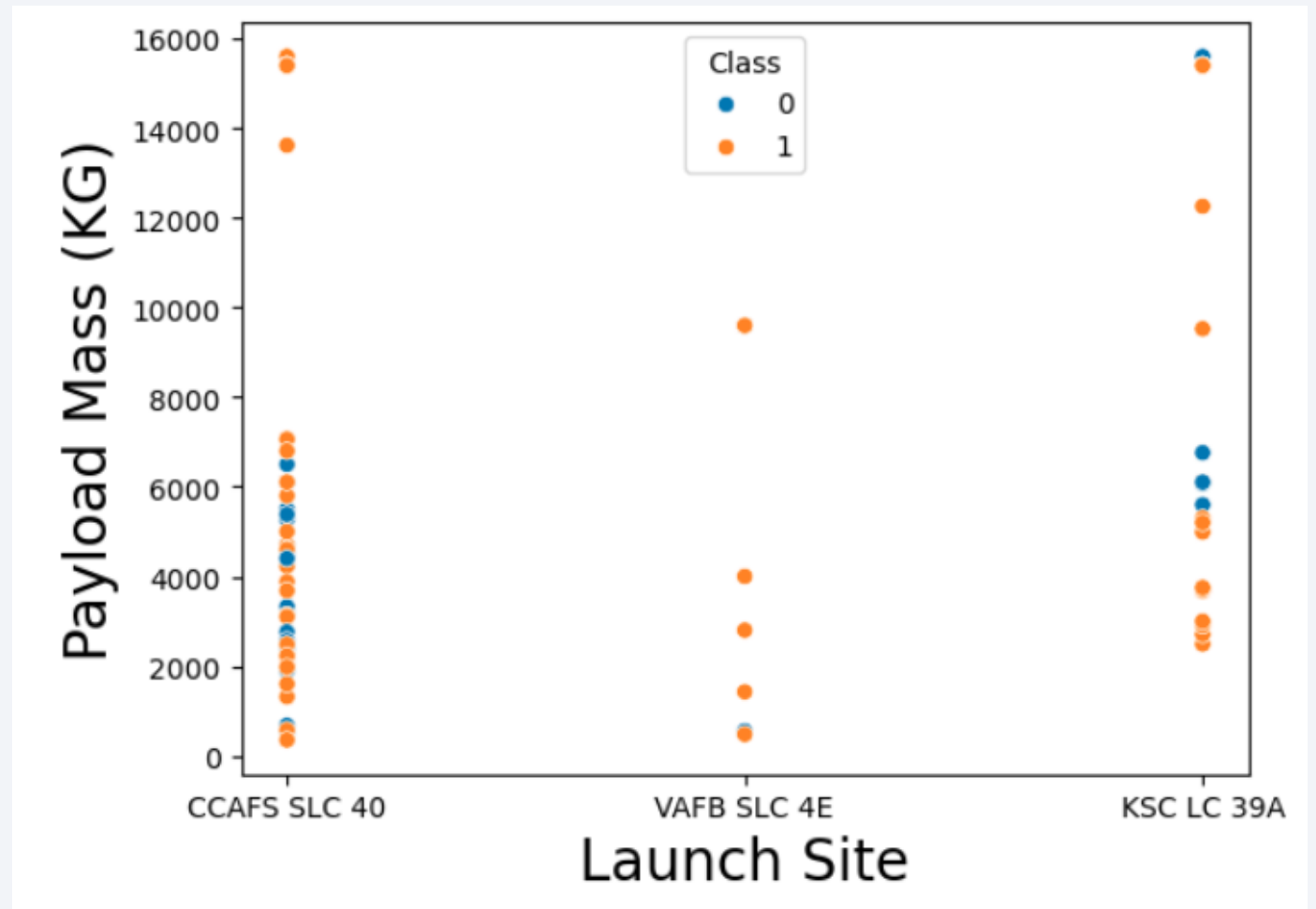
Flight Number vs. Launch Site

The scatter plot is a visual summary of launch outcomes by site over time, making it easy to spot which sites were used most and how mission success evolved.



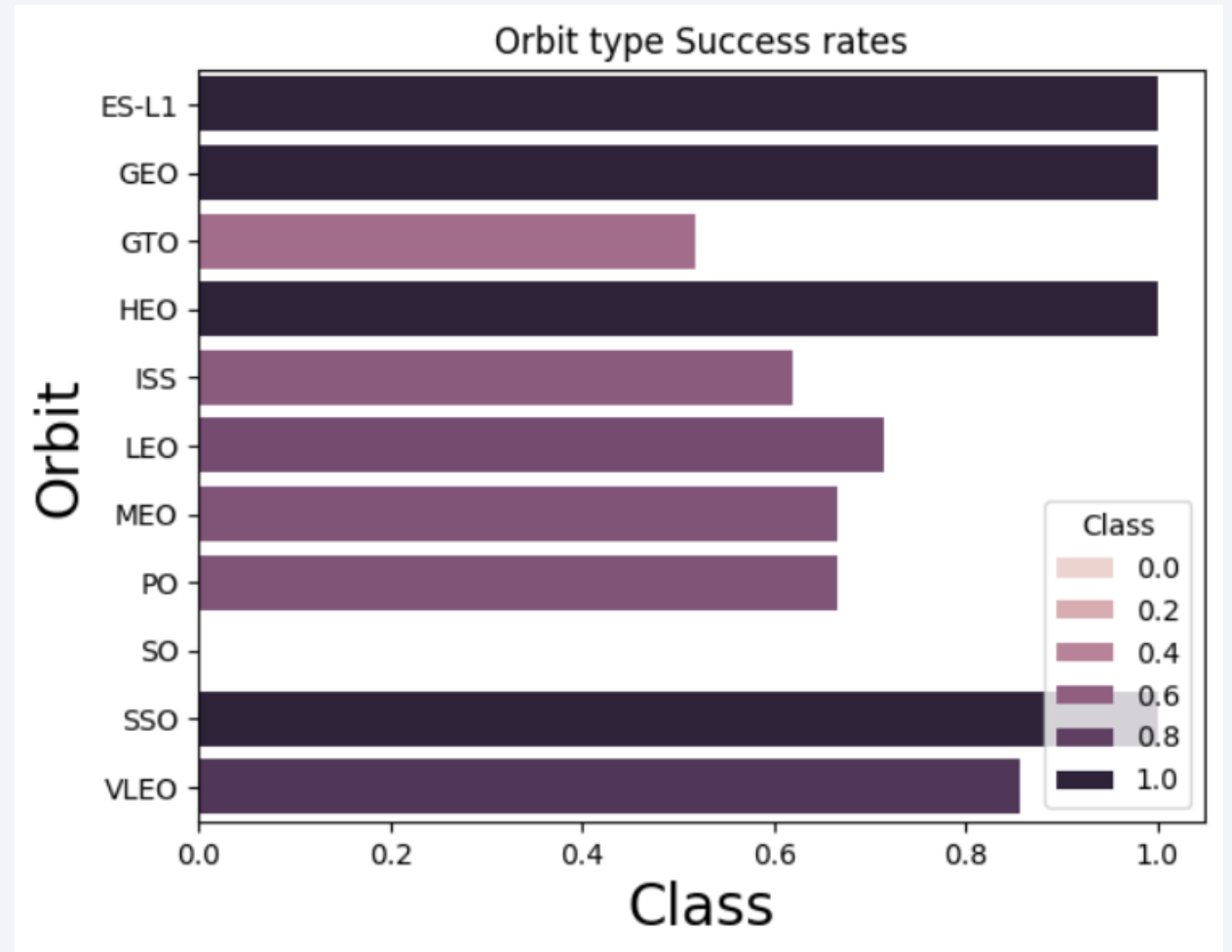
Payload vs. Launch Site

- Launch site CCAFS SLC 40 has the densest cluster of launches in the 0-7000 kg range with a mix of successes and failures,
- while KSC LC 39A handled the heaviest payloads, whose launches were predominantly successful.



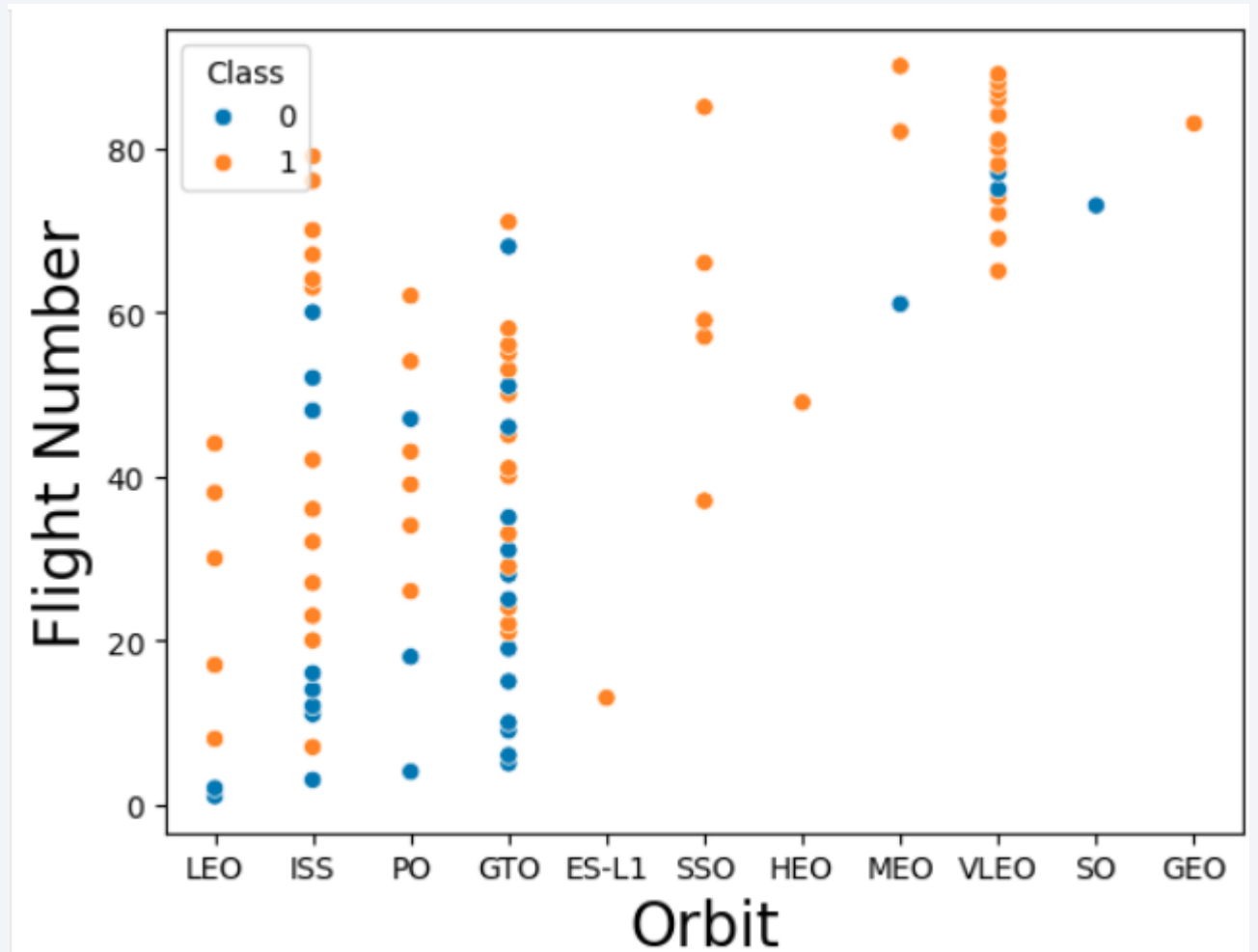
Success Rate vs. Orbit Type

- **ES-L1, GEO, HEO, SSO,** and **SO** all show a **100% success rate**. **GTO** has the lowest success rate at roughly **50%**.
- ISS and LEO sit in the **60–70%** range.



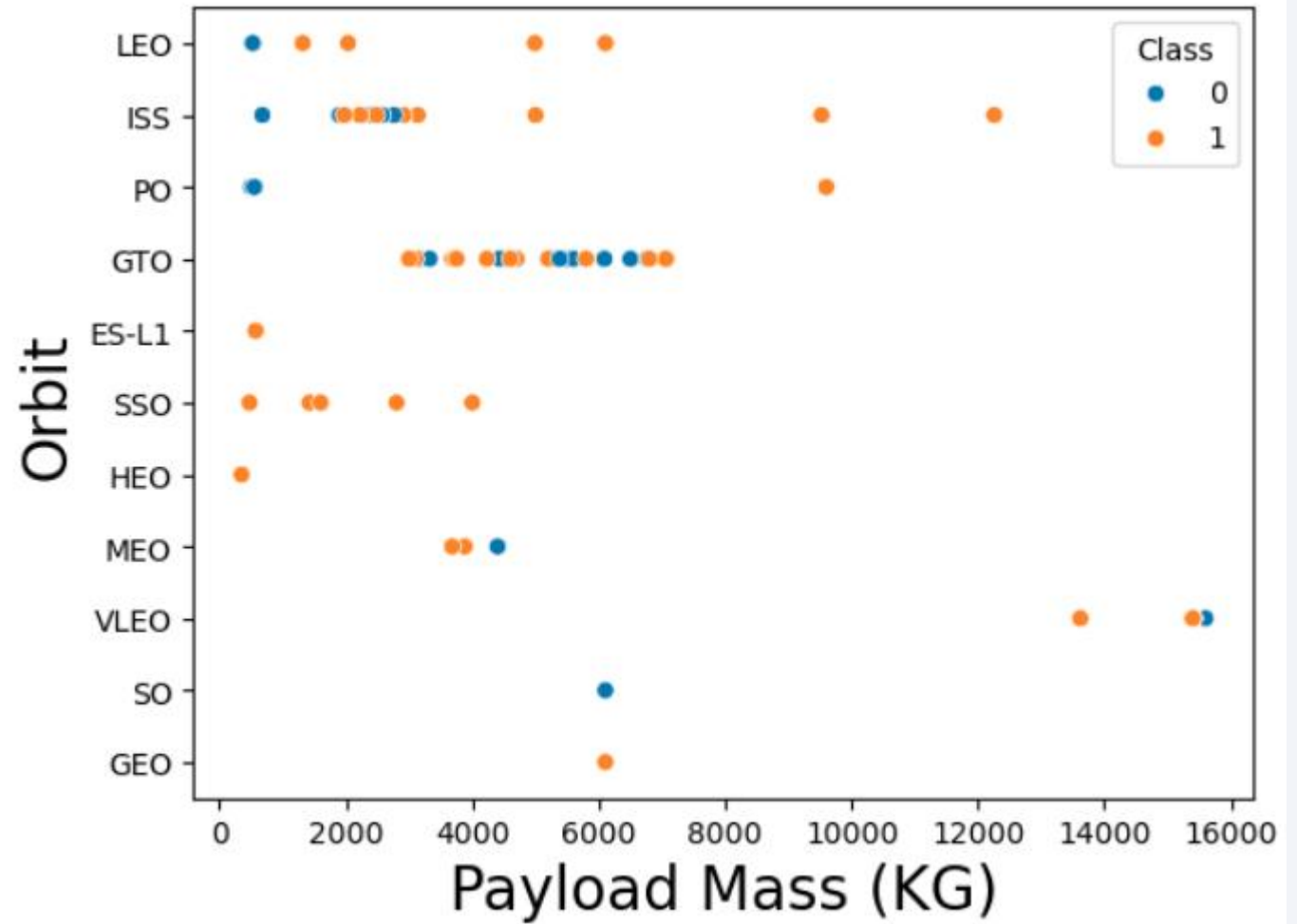
Flight Number vs. Orbit Type

- LEO, ISS, and GTO have clusters of earlier flights.
- SO, VLEO, GEO have branches of later flights, with much higher successes (Class 1)
- Failures are more concentrated in earlier flights across orbit types



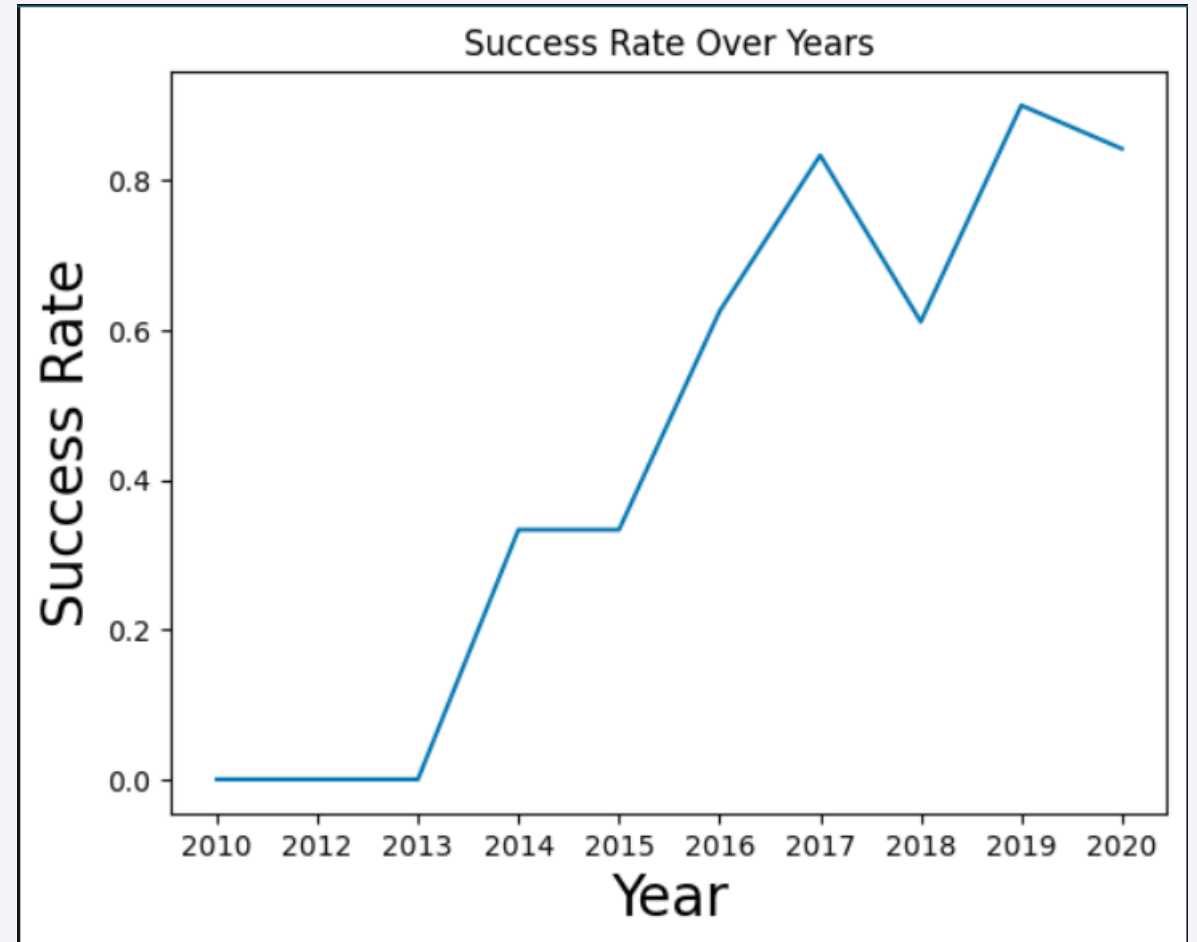
Payload vs. Orbit Type

- GTO missions span a wide payload range (2000-7000KG), with mixed results.
- ISS missions cluster around 2500KG.
- Heaviest payloads are for LEO, VLEO, and ISS orbits.
- ES-L1, HEO, and SSO have few data points.



Launch Success Yearly Trend

- There is an overall increasing success of operations across the years.
- Steady climb from 2013 to 2016 to 60%, then peaking to 90% in 2019
- Notable dip around 2018 to 60% before recovery.



All Launch Site Names

- Find the names of the unique launch sites
- The unique launch sites are CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Early missions show a mix of failures or no landing attempts, consistent with experimental phase.
- F9 v1.0 versions were updated across later events
- Booster recovery wasn't yet part of the program

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- The total payload carried by NASA boosters across all Operations was a grand 619,967 KG

Payload

619967

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- Present your query result with a short explanation here

```
AVG(PAYLOAD_MASS_KG_)
```

```
2928.4
```

First Successful Ground Landing Date

- The earliest successful landing of a SpaceX rocket was 22nd of July, 2018.

MIN(Date)

2018-07-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

- The Booster Names which successfully landed on drone ship masses between 4000 and 6000 are:

- F9 FT B1022
- F9 FT B1026
- F9 FT B1021.2
- F9 FT B1031.2

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- There 99 Successes, one of which having an unclear payload status, and one failure in flight

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Present your query result with a short explanation here

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- Both failed landing outcomes in drone ships Launched from Site CCAFS LC-40
- The differences is that one failed in January (Month 1) with a F9 v1.1 B1012 booster, while the other failed in April (Month 4) with a F9 v1.1 B1015 booster.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- 10 Launches had no landing outcomes, 5 succeeded and failed drone ships, 3 on ground pads.
- 3 were controlled to land on the ocean compared to 2 others, a failed to land with parachute assistants

Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

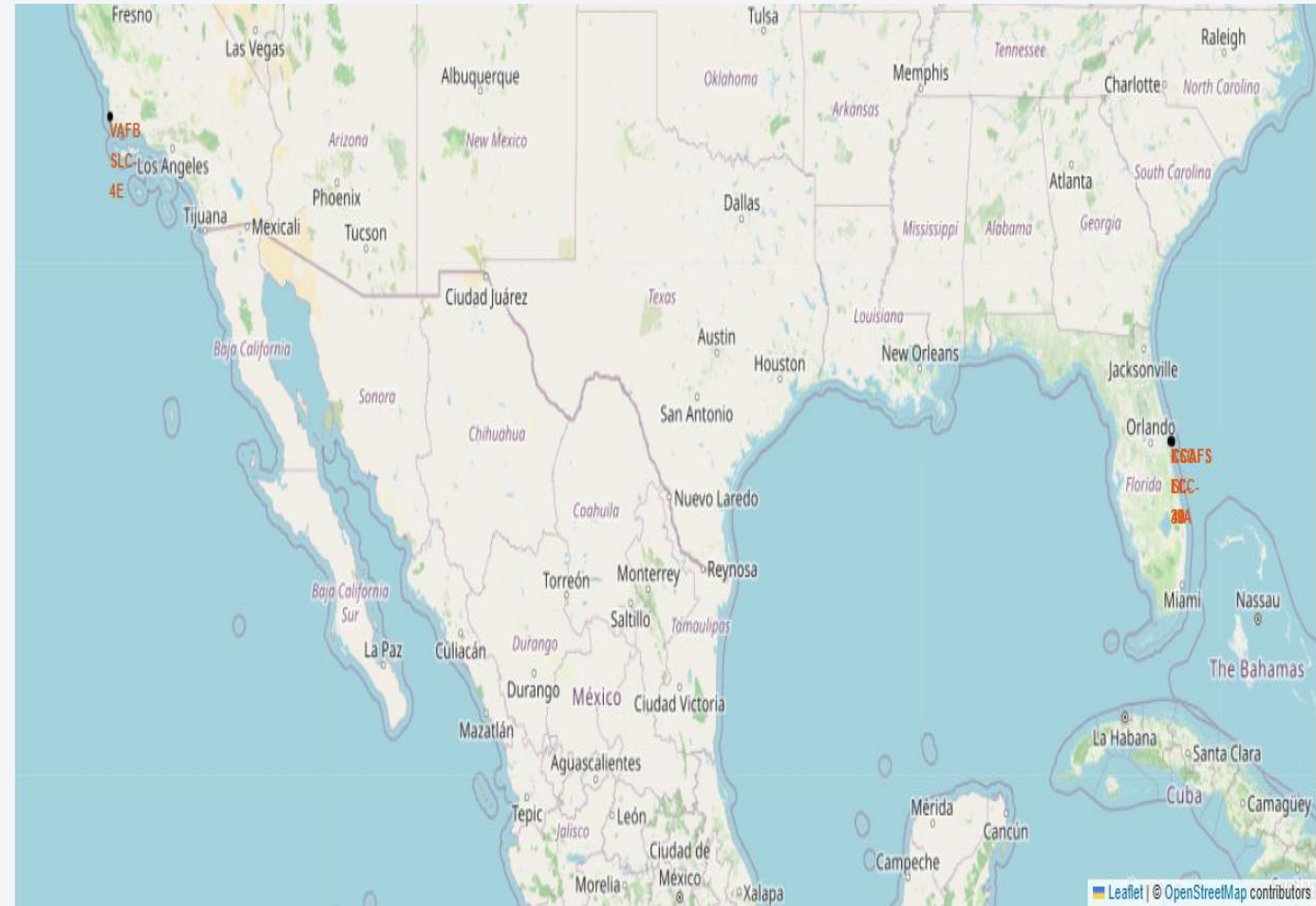
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and cloud patterns. The lights are concentrated in the lower right portion of the image, while the upper left shows a clear blue sky.

Section 3

Launch Sites Proximities Analysis

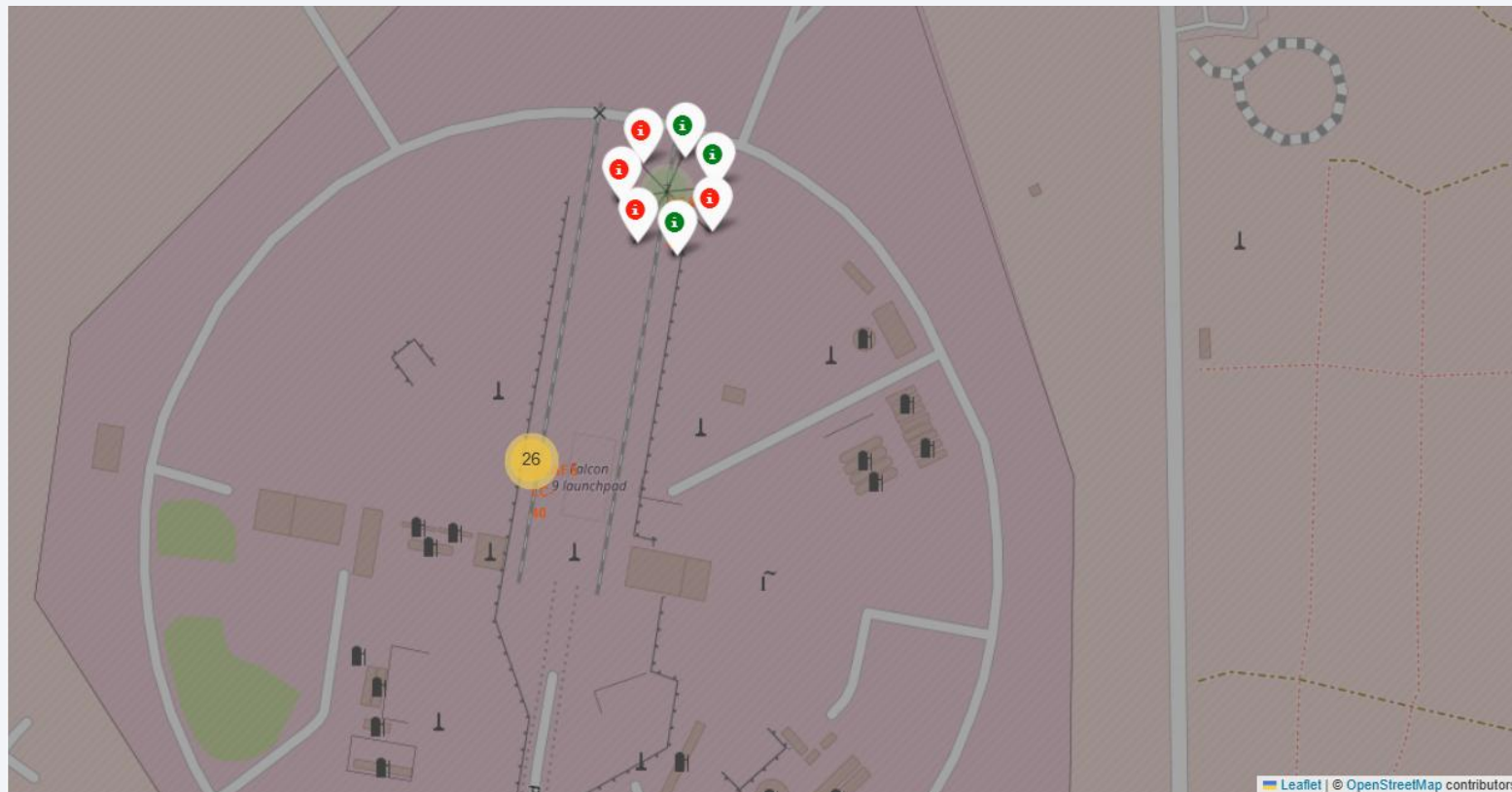
Location Of Launch Sites

- Launch site VAFB SLC-4E is the only one in the US west coast, at California while the rest can be found by the east coast in Orlando.



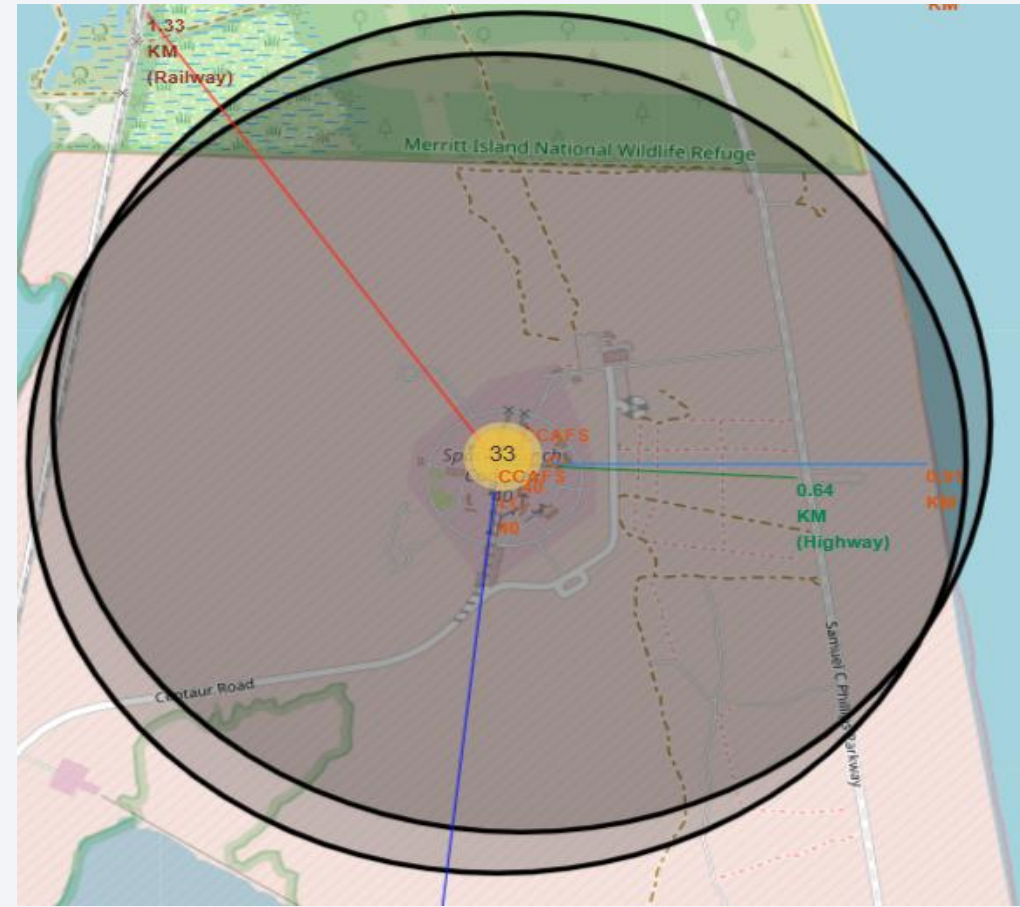
Launch outcomes of Site CCAFS SLC-40

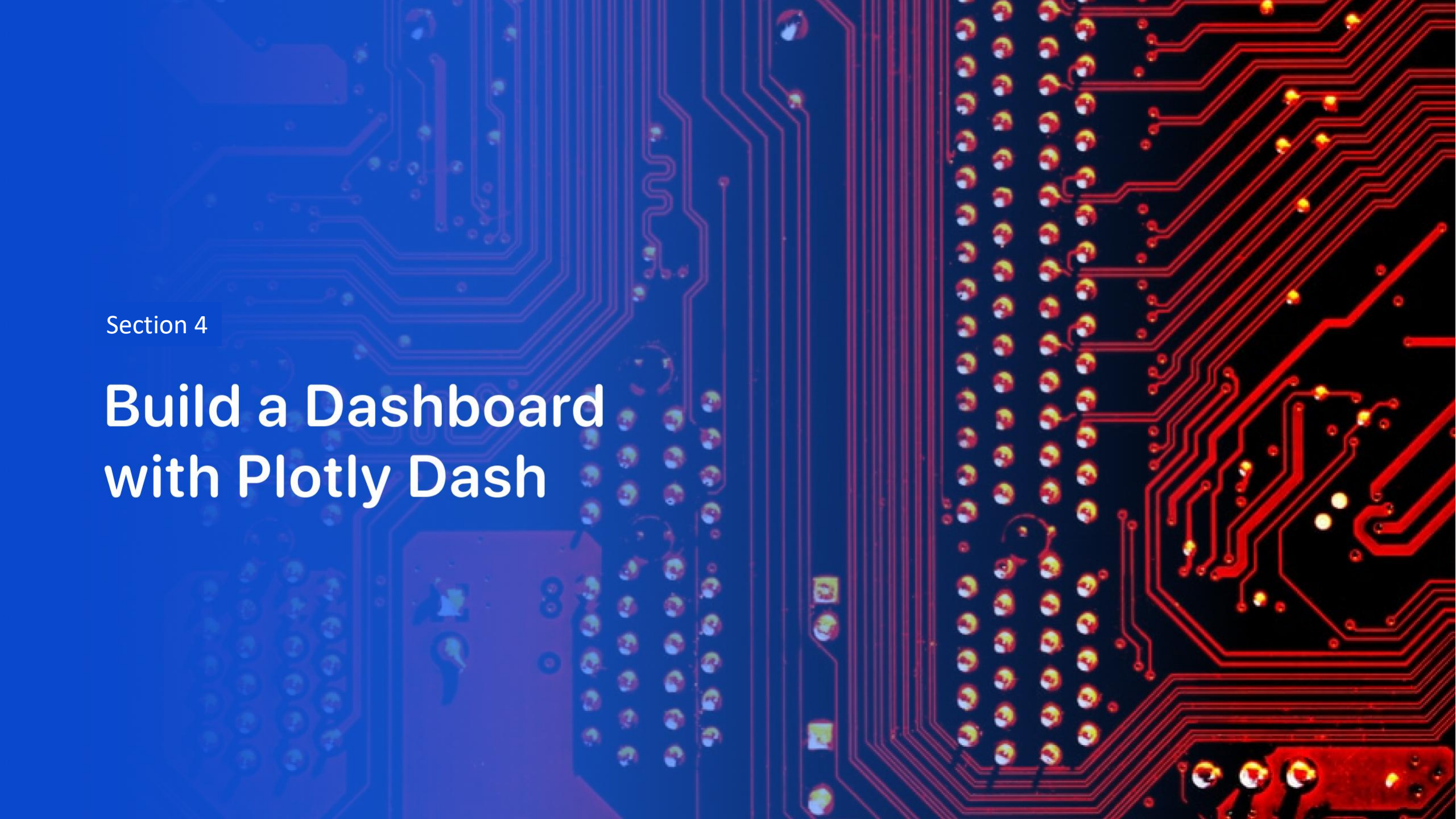
CCAFS SLC-40 is presenting 3 out of 7 successful launches across all records



Launch Site CCAFS LC-40 distances from Highway, Railway, and Coast

- CCAFS LC-40 is .64 Km away from the nearest highway, .91 km away from the coast, and 1.33 Railway



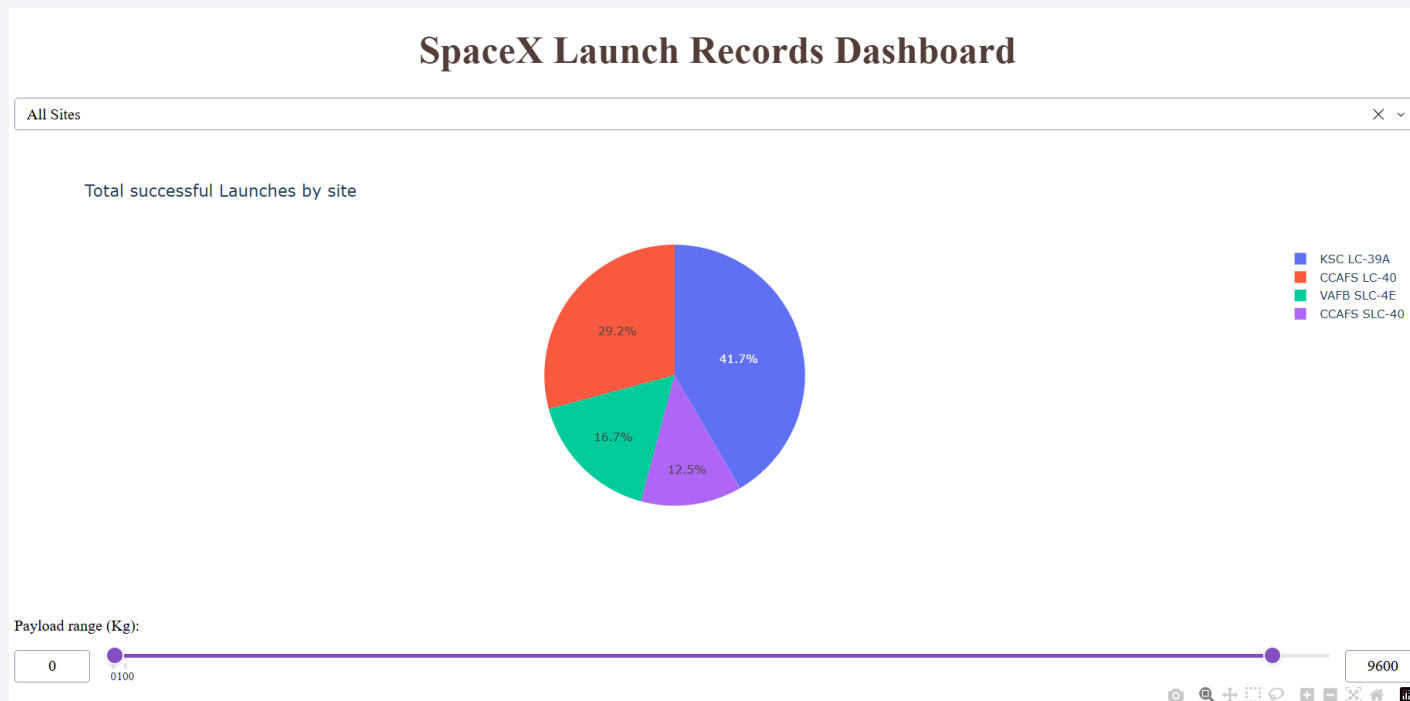


Section 4

Build a Dashboard with Plotly Dash

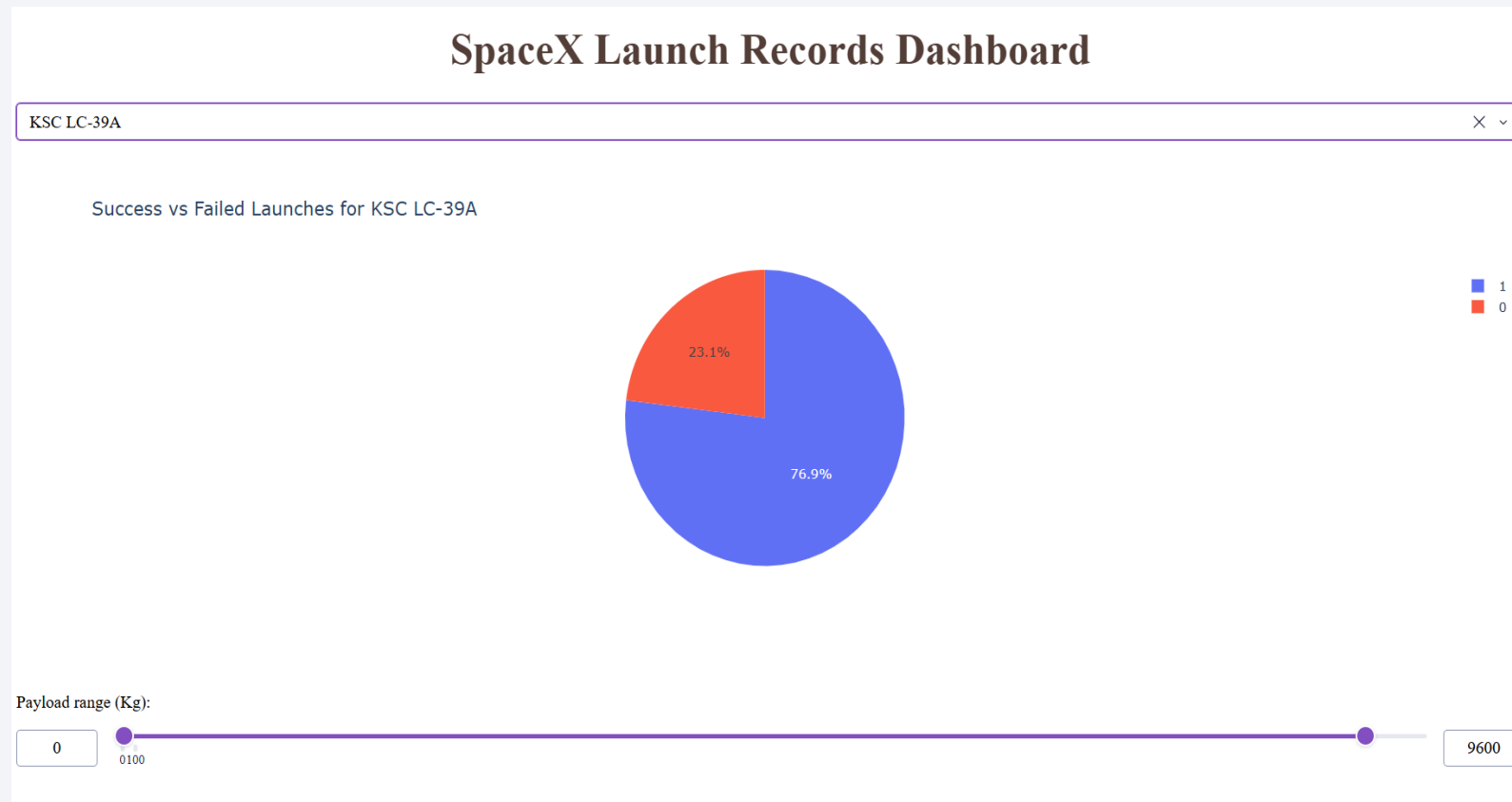
Launch success Count for all sites

- KSC LC-39A is the launch site with the most successes at 10, followed by CCAFS LC-40 by 7.
- The lowest successes came from Launch sites VAFB SLC-4E and CCAFS SCL-40 with 4 and 3 successful launches so far.



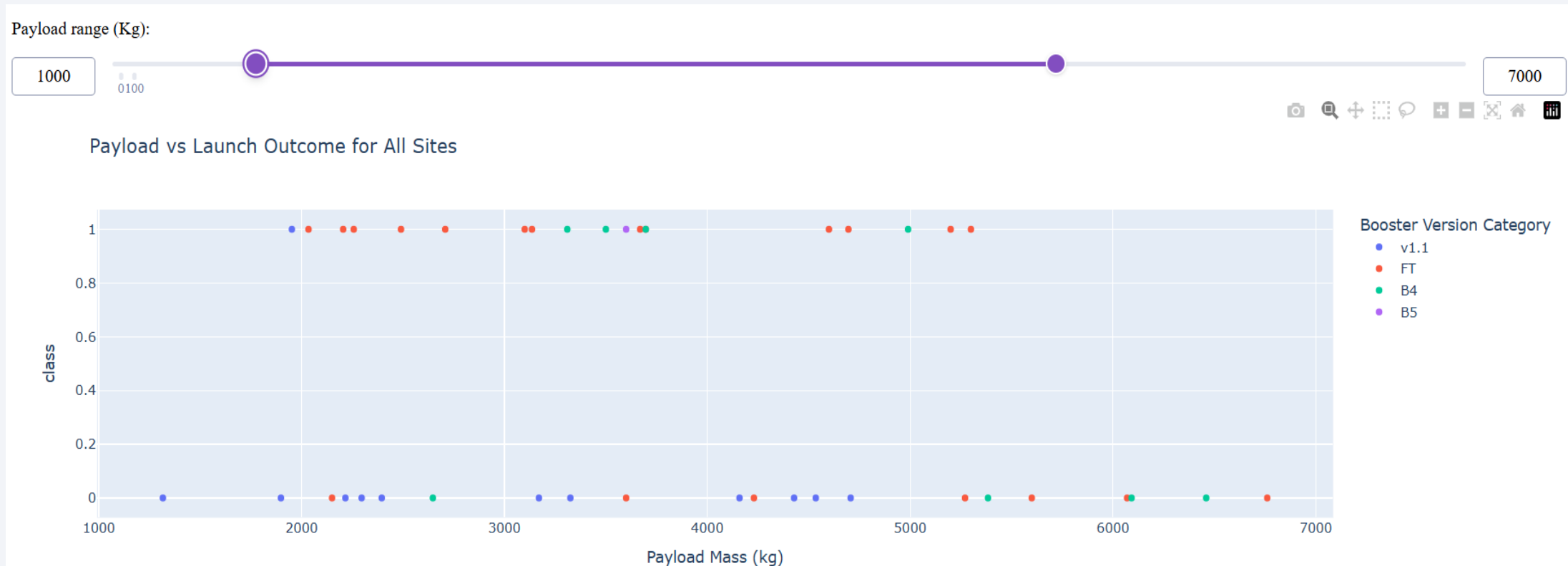
Pie chart of the launch site with the highest success ratio

- Launch site KSC LC-39A has the highest success launch rate of 76.9% (10) to 23.1% (3 failures).



Payload vs. launch outcome scatter plot for all sites with payloads between 1000 and 7000

- The most successful launches occurred in payloads between 2000-5300KG in the selected range with booster version categories FT and B4, with failures spread sparsely.



Section 5

Predictive Analysis (Classification)

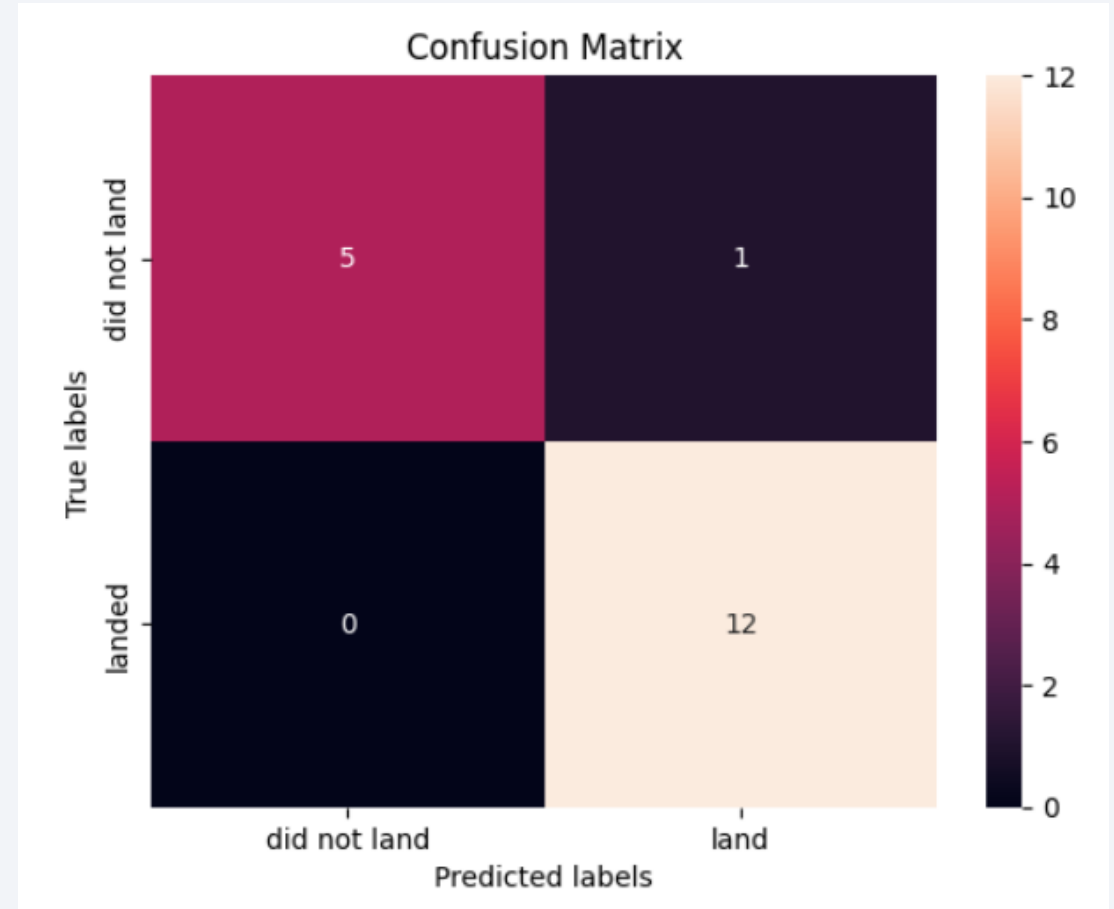
Classification Accuracy

- Decision trees are the best classifier of all trained machine learning models at 94.44% accuracy

Machine Learning Model	Accuracy(%)
Logistic Regression:	0.8333
SVM:	0.8333
Decision Tree:	0.9444
KNN	0.8333

Confusion Matrix

- Decision trees correctly predicted 5 cases where the rocket didn't land and 12 more that did, with only 1 false positive in the former



Conclusions

- Decision Tree is the best predictor at 94.44% accuracy — It significantly outperforms the other three models (Logistic Regression, SVM, and KNN), which all scored 83.33%.
- Landing success rate climbed from near-zero to 90% — Between 2013 and 2019, SpaceX showed a dramatic upward trend, confirming continuous operational improvement.
- KSC LC-39A is the top-performing launch site — It handles the heaviest payloads with the highest success rates, whereas CCAFS SLC-40 shows mixed results across a wider payload range.
- Orbit type is a make-or-break factor — ES-L1, GEO, HEO, SSO, and SO orbits hit 100% landing success, while GTO sits at roughly 50%.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

